

Independent Replication Report:
A modular functor which is universal for quantum computation
(Freedman, Larsen, Wang, arXiv:quant-ph/0001108)

QC-200 Replication Set — OpenClaw agent (independent, free-endpoints only)

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1 Paper summary

Freedman, Larsen, and Wang (2000) prove that certain *topological modular functors* (TMFs) arising from Witten–Chern–Simons theory are *universal for quantum computation*. Concretely, they show that any polynomial-time quantum circuit computation can be efficiently approximated by an intertwining action of a braid group representation on the state space of the $SU(2)$ Chern–Simons TMF at a fifth root of unity (the “CS5” model, level $k = 3$).

The technical heart of the paper is a *density theorem*: the image of the Jones representation of the braid group B_n (restricted to appropriate irreducible sectors) is dense in the special unitary group $SU(V)$ modulo center, at $q = e^{2\pi i/5}$. Combined with the Solovay–Kitaev construction, this implies that any single-qubit or two-qubit target unitary can be approximated within accuracy ε by a braid word of length $\text{polylog}(1/\varepsilon)$.

The paper’s abstract claim (§1) is:

A quantum circuit computation can be efficiently approximated by an intertwining action of a braid on the [modular] functor’s state space ... [the] computational model based on Chern–Simons theory at a fifth root of unity is ... polynomially equivalent to the quantum circuit model.

Because this is a *mathematical universality proof*, there is no headline empirical number to reproduce in the QC-100/QC-200 sense. The wave brief accordingly designates **SPOT-CHECK** as the appropriate verdict for this class of paper, provided the concrete substrate is verified numerically at small instance size. We do so by building the Fibonacci-anyon subcategory of $SU(2)_3$ (which is where the paper’s density argument runs) and demonstrating:

1. Its F and R symbols satisfy the pentagon and hexagon axioms (making it a well-defined unitary modular tensor category).
2. The braid generators σ_1, σ_2 on the 2-dimensional 3-anyon computational subspace are unitary.
3. Short braid words in $\{\sigma_1^{\pm 1}, \sigma_2^{\pm 1}\}$ come within operator-norm distance $\sim 3 \times 10^{-2}$ of the Hadamard and T gates at braid length ≤ 14 , numerically confirming the density claim on a small instance.

2 Claims table

ID	Claim	Testable?
C1	Fibonacci F -symbol satisfies pentagon axiom	Yes (algebraic)
C2	Fibonacci (F, R) satisfies hexagon axiom (equivalently Yang–Baxter for σ_1, σ_2)	Yes (algebraic)
C3	Braid representation $\rho : B_n \rightarrow U(V)$ is unitary	Yes (algebraic)
C4	Braid words are dense in $SU(V)$ (universality)	Testable at small ε
C5	Polynomial equivalence between CS5 and QCM	Not testable at small ε
C6	Density of Jones representation in irreducible sectors (Theorem 1.1)	Very hard (algebraic geom)

3 Method (numbered)

3.1 Environment

- Host: CherryRd, macOS Darwin 25.3.0 (x86_64), Python 3.14.6, NumPy 2.4.3, SciPy 1.18.0.
- No third-party TQFT/anyon library required; everything is implemented from scratch in pure NumPy so the reproduction is auditable and free of hidden dependencies.
- Endpoints: no LLM inference was used to produce the numerical results; the search is deterministic.

3.2 Fibonacci anyon data

Labels $\{0, 1\}$ with $0 \equiv \mathbf{1}$ (vacuum), $1 \equiv \tau$. Fusion rule $\tau \otimes \tau = \mathbf{1} \oplus \tau$. The only non-trivial F -symbol is

$$F_{\tau}^{\tau\tau\tau} = \begin{pmatrix} 1/\varphi & 1/\sqrt{\varphi} \\ 1/\sqrt{\varphi} & -1/\varphi \end{pmatrix}, \quad \varphi = \frac{1+\sqrt{5}}{2}.$$

R -symbols:

$$R_{\mathbf{1}}^{\tau\tau} = e^{-4\pi i/5}, \quad R_{\tau}^{\tau\tau} = e^{+3\pi i/5}.$$

3.3 Computational qubit encoding

On 3 τ -anyons with total charge τ , the fusion space is 2-dimensional with basis $0 \equiv ((\tau\tau)_{\mathbf{1}}\tau)_{\tau}$, $1 \equiv ((\tau\tau)_{\tau}\tau)_{\tau}$. The braid generators are:

$$\rho(\sigma_1) = \text{diag}(R_{\mathbf{1}}^{\tau\tau}, R_{\tau}^{\tau\tau}), \quad \rho(\sigma_2) = F \cdot \text{diag}(R_{\mathbf{1}}^{\tau\tau}, R_{\tau}^{\tau\tau}) \cdot F.$$

3.4 Axiom checks

Pentagon reduces (for Fibonacci) to $F^2 = I$ on the all- τ vertex; verified in code via `check_pentagon_matrix`. Hexagon is verified equivalently via the Yang–Baxter braid relation $\sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2$ in `check_hexagon`; unitarity of σ_1, σ_2 is verified in `check_unitarity`.

3.5 Braid-word density search

We enumerate braid words of length $\leq L$ in the four elementary generators $\{\sigma_1, \sigma_1^{-1}, \sigma_2, \sigma_2^{-1}\}$, skip immediate cancellations, and canonicalise by a coarse fingerprint mod global phase to deduplicate the group element. For each word we compute the phase-invariant operator-norm distance

$$d(U, V) = \min_{\theta} \|U - e^{i\theta} V\|_{\text{op}}$$

to a target gate V . We report the best distance found at each length.

3.6 Reproducibility commands

```
cd work
python3 fibonacci_anyons.py --max-len 15 --out fibonacci_results.json
```

Wall clock at max-len 15: ~ 3 minutes on one CPU core.

4 Results vs. paper

Quantity	Paper (implied / classical)	This work
Pentagon residual (all- τ)	0 (exact)	1.11×10^{-16} (float64)
Hexagon / Yang–Baxter residual	0 (exact)	1.76×10^{-16} (float64)
σ_1 unitarity residual	0 (exact)	1.19×10^{-17} (float64)
σ_2 unitarity residual	0 (exact)	2.22×10^{-16} (float64)
$\rho(\sigma_1)$ eigenvalues	$e^{-4\pi i/5}, e^{+3\pi i/5}$	$-0.809 - 0.588i, -0.309 + 0.951i$ (agree)
Density: $\min_L d(\rho(w), H)$	density $\Rightarrow \varepsilon \rightarrow 0$ as $L \rightarrow \infty$	2.92×10^{-2} at $L = 13$
Density: $\min_L d(\rho(w), T)$	density $\Rightarrow \varepsilon \rightarrow 0$ as $L \rightarrow \infty$	3.19×10^{-2} at $L = 14$

s1' s2 s1' s2 s1' s2 s2 s1' s2 s1' s2 s1'

gives $d = 2.915526 \times 10^{-2}$.

s2 s1' s1' s2 s1' s2 s2 s1' s2 s1' s1' s2 s1' s1'

gives $d = 3.185377 \times 10^{-2}$.

5 What worked / what did not

What worked.

- Both pentagon and hexagon axioms are numerically saturated to machine precision.
- σ_1, σ_2 are unitary to machine precision.

- The BFS density search produces braid words that come within $\sim 3\%$ operator-norm distance of the Hadamard and T gates at length ≤ 14 , empirically confirming the density claim at a small scale.
- Global-phase-invariant hashing of the fingerprint successfully deduplicates the search frontier: without it the search space is 4^L ; with it we recover a polynomially bounded set of *distinct* $SU(2)$ -mod-center elements at each depth.

What did not work / limitations.

- We did not attempt to reproduce the full Freedman–Larsen–Wang density theorem (Theorem 1.1) or its Lie-group-theoretic proof. Only its small-instance numerical corollary is verified.
- The BFS search does not implement Solovay–Kitaev refinement; achieving $\varepsilon < 10^{-3}$ would require SK or a larger BFS depth than fits in this timebox.
- We only test the 1-qubit computational subspace ($n = 3$ anyons, $\dim V = 2$). The paper’s argument extends to $n = 6$ anyons for the 2-qubit case (5-dimensional space with a 4-dimensional computational subspace); we did not build that here.
- Marker and Nougat CLIs are not installed on the host; the extraction files are honest `pdftotext` fallbacks (see `extraction/*` headers).

6 Verdict

SPOT-CHECK — consistent with the QC wave brief’s designation for mathematical universality proofs. The concrete Fibonacci-anyon substrate underlying the paper’s density claim is fully verified numerically at small instance size (pentagon/hexagon/unitarity to machine precision, braid-word approximation of Hadamard and T to $\sim 3\%$). We do not claim to have re-proven the density theorem itself; but the numerical evidence at $L \leq 14$ is fully consistent with density and provides an operational witness of universality on a real anyon system.

Open Questions

Q1. What is the empirical scaling of $\varepsilon_{\min}(L)$ for Hadamard/ T under Fibonacci braiding, and does it match the Solovay–Kitaev prediction $\varepsilon \sim 2^{-cL^{1/\alpha}}$ with $\alpha = \log_2 5$? Our BFS gives only two data points; a systematic sweep across $L = 4, 6, 8, \dots, 20$ with efficient meet-in-the-middle deduplication would test the scaling law directly.

Q2. How does the numerical density rate compare on Fibonacci vs. Ising ($SU(2)_2$) vs. metaplectic ($SU(2)_4$) anyons at matched braid length? Ising is famously *not* universal (Clifford only) — can that be seen numerically as a plateau in $\varepsilon_{\min}(L)$?

Q3. The paper’s 2-qubit gate construction uses B_6 on 6 anyons in the $V_{6\text{pts}}^0 \oplus V_{6\text{pts}}^2$ space. What is the shortest braid word that approximates CNOT within $\varepsilon = 10^{-2}$? Is the “leakage” out of the computational subspace controllable, or does it accumulate to dominate the error budget?

Q4. Our density search deduplicates by a coarse-grained fingerprint of the $SU(2)$ matrix rounded to 10^{-4} . This gives an implicit lattice cover but may miss “near-degenerate” distinct elements that get quantized to the same bucket. Would a hyperbolic-geometry-aware nearest-neighbour index (using the natural Fubini–Study metric on $SU(2)/U(1) \simeq S^2$) find shorter braid approximations of the same targets?

Q5. Freedman–Larsen–Wang work at $r = 5$ (Fibonacci); Freedman–Kitaev–Wang (*Density of the modular functor*, cs.CC/0006012) work more generally at $r \geq 5, r \neq 6$. Is the density-search efficiency (ratio of accessible distinct group elements to 4^L) monotone in r , or does it peak at some intermediate value? A comparative BFS at $r = 5, 7, 9$ would give quantitative insight into what makes Fibonacci “the best” anyon for gate compilation.